NYSEG Smart Home Rate: OptimizEV Smart Charging Pilot Date (3Q2020) Active

Project Start Date: 09/10/2018 **Project End Date:** 04/31/2021

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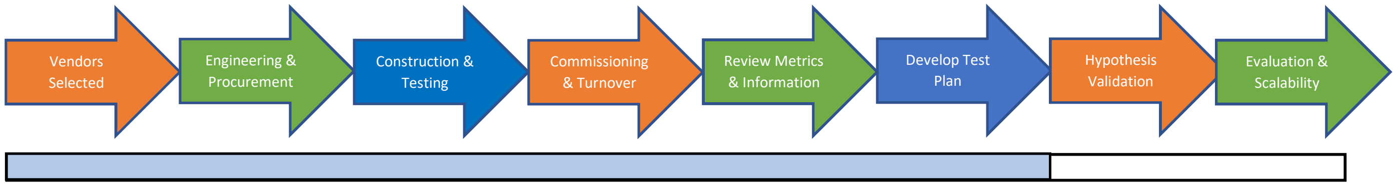

Project Summary: The *OptimizEV Smart Charging pilot* explores the use of deadline-differentiated pricing for Service Class 1 customers with plug-in electric vehicles. It gathers insight into the degree to which customer behaviors can be influenced through indirect (e.g., price) and direct control signals. Ultimately, the ability to efficiently learn an accurate model of customer response characteristics may substantially enhance the ability of the distribution system platform operator to optimize the mix of distribution grid resources.

Lessons learned:

- Little downtime in communications for participants after providing cellular gateway connection instead of using customers' WiFi.
- The volume of sessions has been affected by the COVID-19 pandemic; however, overall, the pilot has seen high instances of optimized charging.

Issues Identified: During a project team meeting in September, questions were drafted up to be answered by future smart charging pilots. Some questions were raised by the vendor, Kitu Systems, such as what the grid benefits are, what is the economic incentive tipping point for a customer to choose to go through the process required to engage in an optimized charging session, what is the minimum participation required of a given population to achieve the desired peak load reduction, and more.

Solutions Identified*:* Identify technical specifications and data requirements early in the project as well as automate data management for all parties.

Recent Milestones/Targets Met:

- June, July, August discounts applied to customer bills
- Determined customer satisfaction over the first six months of the program through a survey
- Project team explored findings from the first six months of project launch

Upcoming Milestones/Targets:

Q4 2020

- Ongoing analysis to understand and remedy causes of energy mis-deliveries
- Develop a survey exploring factors that influence a customer's willingness to use an optimized session
- Conduct a multi-stakeholder project team meeting.

Reforming the Energy Vision

Demonstration Project Q3 2020 Report

Smart Home Rate: OptimizEV Smart Charging Pilot

NYSEG

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1.0 Executive Summary

New York State Electric and Gas Corporation (NYSEG) submits this quarterly report on the progress of the Smart Home Rate Demonstration Project (SHR Project). The SHR Project gathers insight into the degree to which customer behaviors can be influenced through indirect (e.g., price) and direct control signals. The ability to efficiently learn an accurate model of customer response characteristics may substantially enhance the ability of the distribution system platform operator to optimize the mix of distribution grid resources. The SHR Project is located in the AMI footprint within Tompkins County, NY.

In Q3 2020, OptimizEV had a project team meeting to review the program's six months of progress since launch in March. As a result of this meeting, many lessons learned, results and remaining questions and ideas about program scalability were identified. A customer survey was issued at the end of August and results are in Section 2.1.

Upcoming milestone for Q4 2020 include:

- Design and release a discount-focused customer survey to be conducted in Q4

The following report provides a progress update on the tasks, milestones, checkpoints, and lessons learned in Q3 2020.

2.0 Demonstration Highlights

2.1 Activity Overview

Major completed milestones in Q3 2020 included:

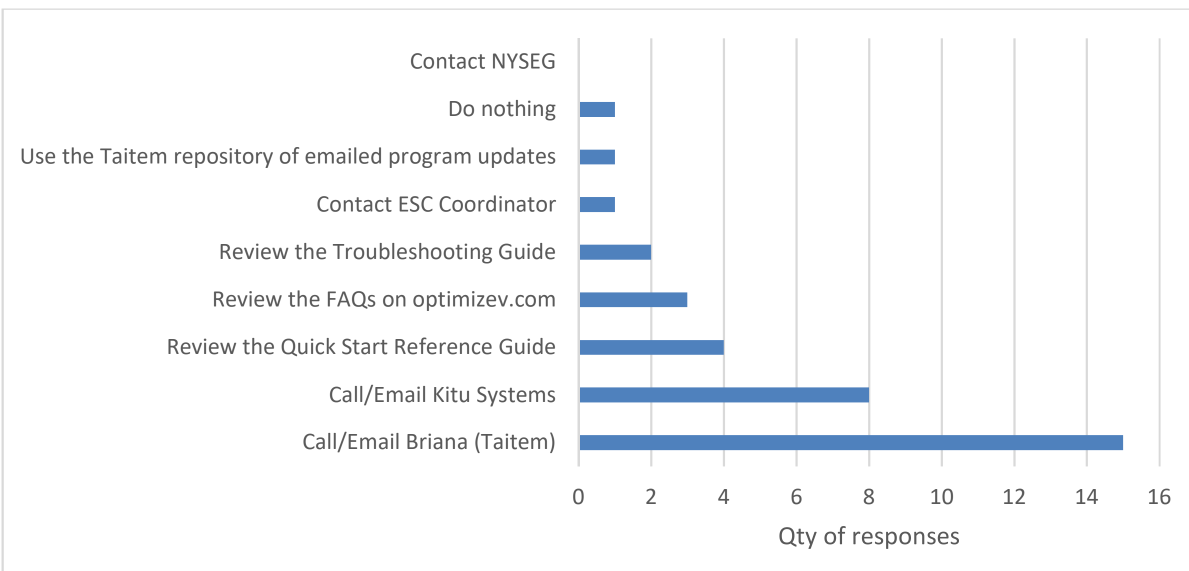
- June, July, August discounts applied to customer bills
- Determined customer satisfaction over the first six months of the program through a survey
- Project team explored findings from the first six months of project launch

2.1.1 August Survey Results

The participant survey emailed on August 12, 2020 had a 53% response rate. The survey contained eight questions about the program, specifically about customer support satisfaction since the program launch in March and decisions around initiating charging sessions.

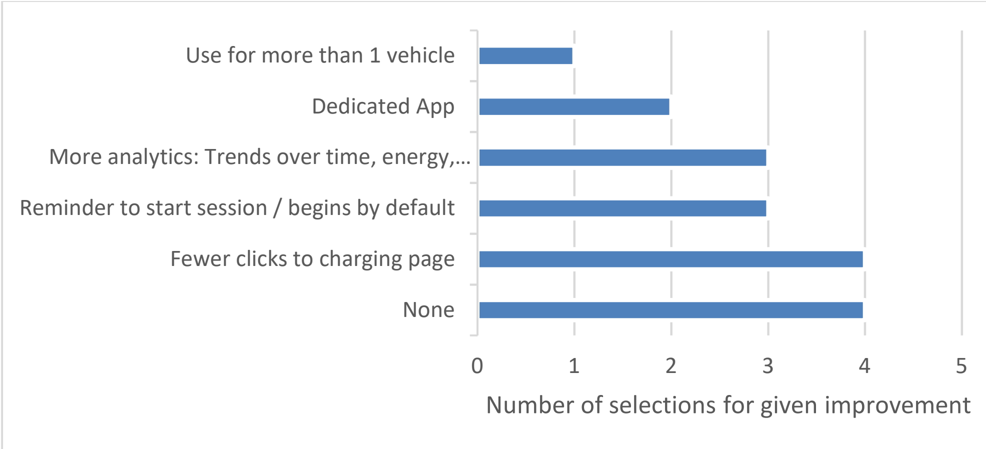
Most respondents indicated that they are satisfied with the available troubleshooting methods (86% of respondents). The most common methods for resolving an issue with the charger or program are contacting Taitem Engineering or Kitu Systems as seen in Figure 1.

Figure 1: Question - If you've had an issue with the charger or program, did you... (check all that apply)



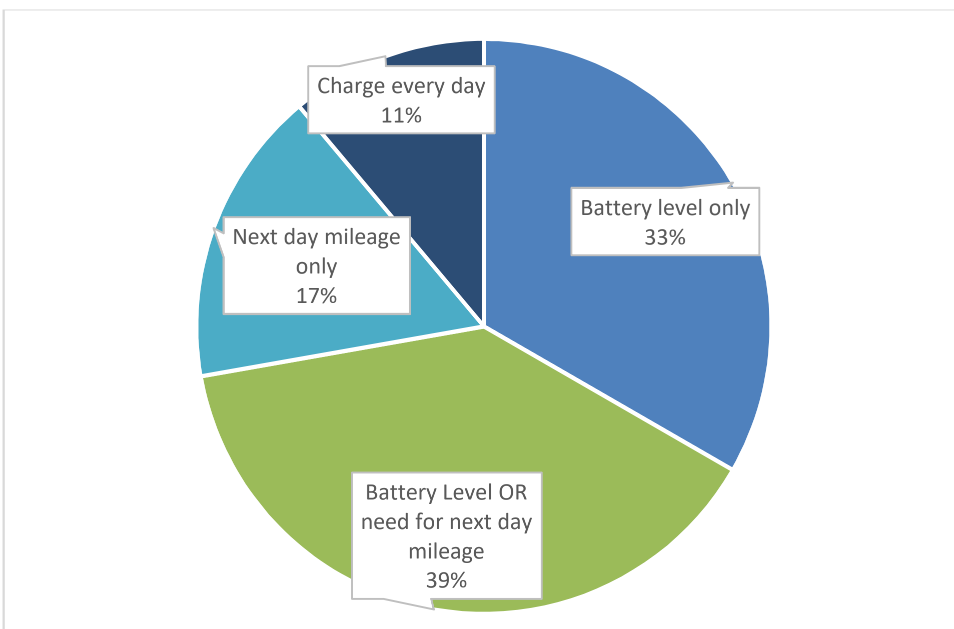
Survey respondents were asked which improvements they would like to see to how their charging sessions are managed. While many indicated that none of the listed improvements were needed, the same number of respondents answered they would like to reach the charging webpage to initiate a session in fewer clicks (Figure 2).

Figure 2: What changes would you like to see to the website and/or how you manage a charging session? Check all that apply.



The survey had questions related to customer interactions with the charging hardware. All respondents used their Level 2 charger instead of their Level 1 charger for their regular charging. Only one respondent reported that they always plug in their car upon arriving home. Customers were also asked about making the decision to charge their vehicle to enhance the project team's understanding of the session data (Figure 3).

Figure 3: Question - How do you decide it's time to charge your EV?



If their decision to charge was based on the battery's state of charge, most respondents wait until the battery has around 30-40% charge left, while one respondent reported waiting until the battery level was under 20% and another respondent reported waiting until under their battery is at half capacity.

2.1.2 Project team meeting

The project team had a large virtual meeting with participants from NYSERDA, Avangrid, Taitem, Cornell University, and Kitu Systems to review the entire project from each of their perspectives. Expectations, data, customer support, and lessons learned were discussed.

For the first six months of the live pilot, the project has shown that it is possible to have granular, to the minute, smart charging controls executed for both for hybrid and all battery EVs. The pilot has also demonstrated an ability to reduce peak loads through the smart charging system. Four specific year/make/model of EVs have not responded to these smart charging controls. How the controls can be adapted for these vehicles is still unknown but is actively explored.

No system software changes that would test customer acceptance of fewer controls, a different baseline load curve, or adjusted discount have been deployed at this time since the data has not been in a steady state due to the ongoing pandemic. The project team uses the following dates in Table 1 to provide context to the data collected through Q3.

Table 1: COVID-19 dates for the Southern Tier

Date	Event
8-Mar	50% working from home
15-Mar	75% working from home
22-Mar	100% working from home
24-May	Phase 1: Businesses begin to reopen
31-May	Phase 2
12-Jun	Phase 3
26-Jun	Phase 4
23-Aug	Cornell prepares for in-person classes

The project team discussed that the user experience is hampered by the customer's inability to see the battery's state of charge. This issue is due to current technology limitations. Not all EVs are able to report to the charger or to the owner its state of charge. Furthermore, not all EVs provide a value of battery level as a percentage, which would match the units in the OptimizEV portal where a customer indicates battery level to execute a smart charging session.

Although the monthly discounts are relatively low for an average NYSEG customer, there has been a strong adoption of optimized charging. However, since the grid benefits are not tied into the discount in this program, it is possible that customers in a future program could receive higher discounts. On the individual level, the project team has seen that the cumulative monthly discounts tend to be around $4-$5, which is close to 10% of the average Service Class 1 customer delivery bill in NYSEG territory.

The scalability of a utility EV program the same as or similar to OptimizEV was discussed. In its current configuration, the hardware/software is relatively expensive since participants receive a free charger and a free cellular gateway with no monthly service fee. The controls software provides by-the-minute controls that generate a high volume of data traffic that stresses both the devices and the cloud computing systems. The project team discussed that the number of controls could be reduced with a change in the algorithm. If/when a steady state occurs in the customer behavior currently affected by the pandemic, this could be tested in OptimizEV. This pilot used only one brand/model of charger, which is now discontinued. A future pilot would benefit from testing multiple charger brands thereby allowing participants with more options.

The team discussed the benefits of extending the pilot for additional months to mitigate impacts of the pandemic on charging behaviors, to produce a larger data sample and to alter the algorithm to test impacts including the effects of using fewer controls.

Remaining questions the project team has yet to answer that should be answered by future programs were also discussed and is summarized in Section 4.0.

2.2 Metrics and Checkpoints

This section contains the checkpoint table and Q3 metrics.

2.1.1 Checkpoint table

The checkpoint table remains unchanged since the Q2 report.

Table 2: Checkpoints

Checkpoint	Description
Customer Participation (Test Statement 1, Metric 1)	Measure: Number of program participants How and When: Monthly, using data from Simple Energy and Kitu Expected Target: 10% of total target population, or 35 participants. Target is based on 10% of approximately 353 EV owners within Tompkins Co that also have a smart meter (NYSERDA EValuateNY, August 2018. Selected Municipalities: Ithaca, Lansing, Dryden, Groton, Etna). Solution/Strategy if expectations are not met: Evaluate engagement and communication strategies and evaluate/establish optimal advertising mix.

	<u>Status</u>: Achieved.
Customer Transactions (Test Statement 1, Metric 2)	<u>Measure</u>: Number of charging sessions (transactions) <u>How and When</u>: Monthly, using data from Kitu <u>Expected Target</u>: 425 transactions each month. <u>Solution/Strategy if expectations are not met</u>: Evaluate pricing structure and solicit customer feedback to understand user experience
Customer Engagement (Test Statement 1, Metric 3)	<u>Measure</u>: Percentage of deadline differentiated pricing transactions (i.e. how often a customer is willing to delay their charge for a financial incentive) <u>How and When</u>: Monthly, (number of deadline differentiated charging sessions / total charging sessions) * 100 <u>Target:</u> 50% of all charging sessions use the deadline differentiated pricing at three months, six months, and by the end of the pilot. <u>Solution/Strategy if not met:</u> Evaluate and add/change offerings; evaluate and change price structure; conduct customer survey to obtain direct suggestion/feedback; optimize marketing/PR channels. Send alerts via platform.
Customer Flexibility (Test Statement 2, Metric 1)	<u>Measure:</u> Monthly average slack per customer. <u>How and When:</u> Monthly. For all customers engaging in an optimized session: *Slack* is the minimum time technically required to charge the vehicle subtracted from the delay (hours) the customer chooses. Average slack is slack divided by the number of sessions analyzed. The minimum time required to charge the vehicle is based on the rated power of EVSE and the maximum power the EV can accept. $$\frac{\sum_{j=1}^{m}\left[\sum_{i=1}^{n}\left(d - dmin(E)_{ji}\right)\right]}{total\ number\ of\ optmized\ sessions}$$ Where nj is the number of sessions per customer within month m is the number of customers d is delay chosen by customer (hours) dmin(E) is the minimum amount of time required to charge the vehicle to the customer's request E is energy requested by customer (kwh) <u>Target</u>: 8 hours

	Solution/Strategy if not met: Evaluate and change price structure; conduct customer survey to obtain direct suggestion/feedback; provide deadline differentiated delay charging reminders by email or via the customer online platform by Kitu.

System Efficiency (Test Statement 2, Metric 2)	Measure: Percentage of effective peak reduction (EVSE load of pilot participants only) How and When: Quarterly. (Max power available – Max power delivered) / Max Power Available, where power is in kW, or $\frac{\sum_{j=1}^{m}\left[\sum_{i=1}^{n}(EVSEMaxPowerAvailable_j - EVSEMaxPowerDelivered_{ji})\right]}{\sum_{j=1}^{m}\left[\sum_{i=1}^{n} EVSEMaxPowerAvailable_j\right]} * 100$ Where n_j is the number of sessions per customer m is the number of customers “EVSE Max Power Available” is whichever amount is smaller: the rated power of the EVSE or the maximum power the EV can accept “EVSE Max Power Delivered” is the peak power output of the EVSE during a session Targets: 14% - BEVs, 28% - HEVs Solution/Strategy if not met: Evaluate and change price structure and algorithm parameters. Consider EVSE platform alert to remind customers of incentives. Note: This metric considers individual charger load reduction, not aggregate load reduction.
Discount Claimed (Test Statement 3, Metric 1)	Measure: Average savings ($) per customer How and When: Monthly data from Kitu $\frac{\sum_{j=1}^{m} Savings_j}{m}$ Where m is the number of customers Target: $1.75 per customer Solution/Strategy if not met: Evaluate and change price structure; conduct customer survey to obtain direct suggestion/feedback; provide deadline differentiated delay charging reminders by email or EVSE platform alert.

2.1.2 Metrics Status

The metrics, targets and OptimizEV program were formulated under a set of circumstances and assumptions before the pandemic. The values of the reported metrics are considered inconclusive until regular driving patterns emerge during the post-pandemic months. New patterns showing typical behavior will emerge and at that time more accurate comparisons will be able to be made.

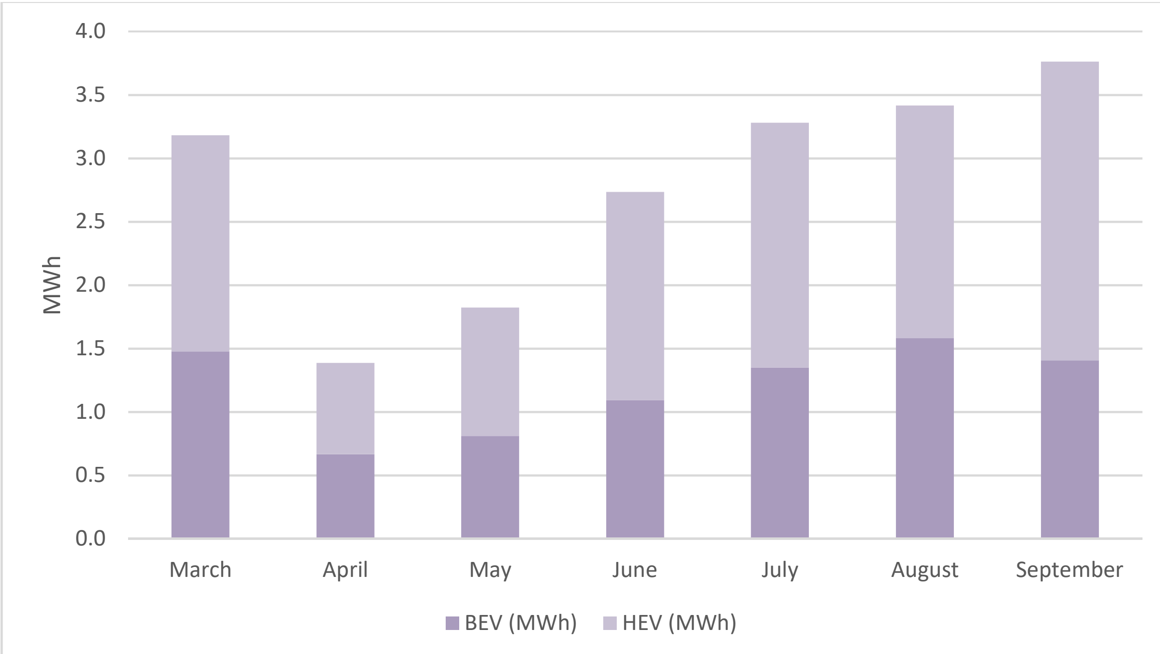
The COVID-19 pandemic has affected the volume of sessions. The number of customer transactions has not returned to values from Q1 2020. However, as seen in Table 3, the number of transactions in September were the highest since March. Additionally, there was a steady increase over Q3 in the average discount claimed per participant, surpassing the target amount in August. The pilot continues to see high instances of optimized charging as seen in the customer engagement metric.

Table 3: Metrics Results for Customer Engagement, Flexibility, and System Efficiency, Transactions and Discount Claimed

Metric	Monthly Target	Jul	Aug	Sep
Customer Engagement	50%	71%	59%	62%
Customer Flexibility	8 hours	8.9 hours	9.4 hours	9.3 hours
System Efficiency BEVs	14%	21%	17%	13%
System Efficiency HEVs	28%	42%	42%	43%
Customer Transactions	425 sessions	271	284	329
Average Discount Claimed	$1.75	$1.70	$1.96	$2.12

Although the number of transactions in September was lower than in March, the electricity consumed (MWh) was higher in September than in March. This means that although there were fewer charging sessions overall, more electricity was consumed per session in September than in March as seen in Figure 4. Additionally, with many customers working from home, their increased flexibility to leave their EVs plugged in provides more flexibility to the grid and therefore has resulted in more dollar savings for them.

Figure 4: Monthly electricity consumption



2.3 Issues

2.3.1 EV not accepting charge during optimized sessions

There are six vehicles that likely do not accept charge due to a sleeping state. One OEM has confirmed that a software update should now fix previous issues with allowing a pilot control signal to keep their EVs awake and ready to accept charging controls. The project team will consider timing, capacity, and quality assurance in Q4 before deciding to deploy the pilot signal for all vehicles.

We continue to monitor any unusual activity resulting in EVs not accepting control signals.

3.0 Work Plan, Budget Review, Upcoming Activities

3.1 Budget Review

[redacted]

[redacted]

3.2 Work Plan

The work plan has not changed since the Q1 2020 report.

Table 5: Work Plan

Milestones / Deliverables	Completion Date
PROJECT PHASES	
P1. Procurement Planning & Design	**9/10/2018**
Implementation Plan is sent to PSC	4/30/2018
Identify EVSE customer platform requirements	4/30/2018
Specify integrations between EVSE customer platform and other Avangrid systems	5/15/2018
Define requirements for selling chargers as a product within NYSEG Smart Solutions	5/15/2018
Release RFP for software / hardware procurement	5/16/2018
Establish EVSE rebate w/NYSERDA	9/10/2018
P2. Build	**12/15/2019**
Algorithm developed	8/15/2019
Procure EVSE customer platform with: • Algorithm from Cornell (discounts, coordinated charging optimization) • Customer portal	12/11/2018
Vendor Kickoff	5/10/2019
Complete survey to support pricing options and User Interface	9/30/2019
Enrollment available on Marketplace	8/30/2019
User Interface customized with survey results	12/15/2019
Develop and integrate EVSE software platform	12/15/2019
Develop targeted marketing plan for pilot	6/30/2019
Launch targeted marketing plan for pilot	7/15/2019
P3. Testing	**2/15/2020**
EVSE platform testing	2/7/2020

Test User Interface	2/7/2020
Verify billing adjustment content	2/15/2020
P4. Business Process Procedures & Training	**7/15/2019**
Customer service training	Ongoing from 7/15/2019
P5. Production	**3/3/2020**
Baseline Data Collection Launch (no optimization)	11/1/2019
Complete required modifications	2/29/2020
Optimized Charging Launch	3/3/2020
Data collection ends	2/28/2021
P6. Evaluation	**4/30/2021**
Submit quarterly reports	Ongoing from Q1 2020
Complete final report with findings and recommendations	4/30/2021

3.3 Next Quarter Planned Activities

In Q4, 2020 the project team will continue with the following tasks:

- Ongoing analysis to understand and remedy causes of energy mis-deliveries
- Develop a survey exploring factors that influence a customer's willingness to use an optimized session
- Conduct a multi-stakeholder project team meeting.

4.0 Recommendations and Lessons Learned

During the project team meeting in September, a list of questions to be answered by future smart charging pilots was raised by Kitu Systems:

- What is the economic incentive tipping point for a customer to choose to go through the process required to engage in an optimized charging session?
- What are the grid benefits, including locational considerations to the utility? The cost/benefit analysis could be further developed.
- If the control software sent fewer controls to the EVSE/EV, the cost to facilitate the program would reduce but would the desired peak load impact be achieved?
- What happens if the control software uses customer WiFi instead of a cellular gateway? Doing so would reduce cost but may also reduce the number of customers who are able to participate due to connection reliability concerns.
- What is the minimum participation required of a given population to achieve the desired peak load reduction?

Recommendations to similar and future projects are as follows:

- Identify technical specifications and data requirements early in the project
- Data is the common requirement of all stakeholders. Automate data management for all parties. Specifically, it’s recommended to have a platform that would provide access to data history (sessions, user, vehicle, platform information) with the ability to sort, organize and extract reports in spreadsheet (editable) format

Additional recommendations for website/application features from the August survey are in Section 2.1.